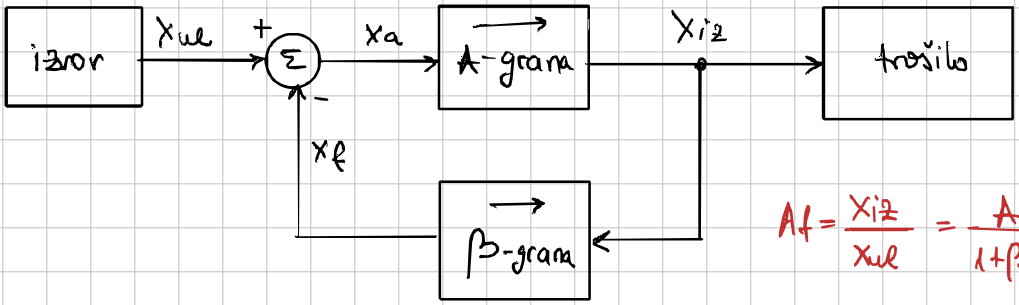
STABILNOST POJAČALA S P.V. VEZOM



$$A_f = \frac{x_{iz}}{x_{ul}} = \frac{A}{1+\beta A}$$

*iako nema izvora, imamo signal, kako?

Frekvencijski odziv pojačala s P.V.

ZA šire područje frekvencija: $A_f(j\omega) = \dfrac{A(j\omega)}{1+\beta(j\omega)\cdot A(j\omega)}$

pojačanje u petlji P.V: $\beta(j\omega)\cdot A(j\omega) = |\beta(j\omega)\cdot A(j\omega)|\cdot \exp(j\varphi_{\beta A})$

↳ način kako se mijenjaju amplituda i faza $\varphi_{\beta A}$ s frekvencijom odredituju stabilnost pojačala s P.V

Specijalni slučaj ω_{180} fazni pomak jednak $\varphi_{\beta A} = \pm 180°$

$\beta(j\omega_{180})\cdot A(j\omega_{180}) < 0,\ \in \mathbb{R} \longrightarrow$ P.V. postaje >0

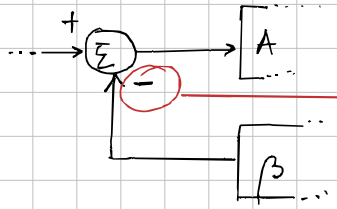
$\boxed{\text{Za } |\beta(j\omega_{180})\cdot A(j\omega_{180})| < 1} \longrightarrow \underbrace{|A_f(j\omega_{180})|}_{\text{konačno pj.}} > |A(j\omega_{180})|$ ⇒ STABILNO

$\boxed{\text{Za } \beta(j\omega_{180})\cdot A(j\omega_{180}) = -1} \longrightarrow |A_f(j\omega_{180})| = \infty$

↳ x_{iz} konačno uz $x_{ul} = 0$

↳ oscilacije

NESTABILNO



minus – trebamo pomak 180° u obje grane (odnosno + i 360°)

↳ $A_f = \dfrac{A}{1-A\beta}$

$\boxed{\text{Za } |\beta(j\omega_{180})\cdot A(j\omega_{180})| > 1}$ → oscilacije se pojačavaju ⇒ NESTABILNO

↳ ugađanjem ovako postižemo nestabilnost

Polovi prijenosne fje pojačala s P.V.

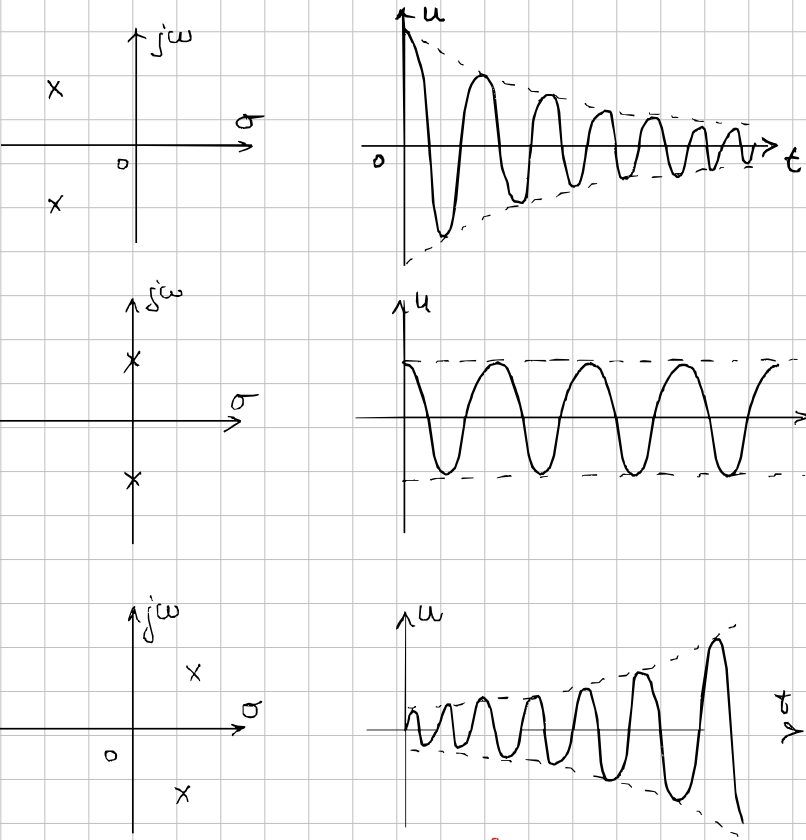
$$A_f(s) = \frac{A(s)}{1+\beta(s)\cdot A(s)}$$ } prijenosna fja poj. s. P.V.

NULE = br. POLOVI = nazivnik (elektro i sis)

$s = \sigma_i \pm j\omega_i$

→ odziv na poremećaj ili smetnju: $u(t) = U_1 \exp(\sigma_1 t)\left[\exp(j\omega_i t) + \exp(-j\omega_i t)\right]$

$u(t) = 2U_1 \exp(\sigma_1 t)\cos(\omega_i t)$



pojačavanje oscilacija

Pojačalo je stabilno ako su realni dijelovi polova negativni ili ako su polovi smješteni u lijevoj poluravnini

Pojačalo s jednim polom u prijenosnoj funkciji

$$A(s) = \frac{A_0}{1+\frac{s}{\omega_p}}$$

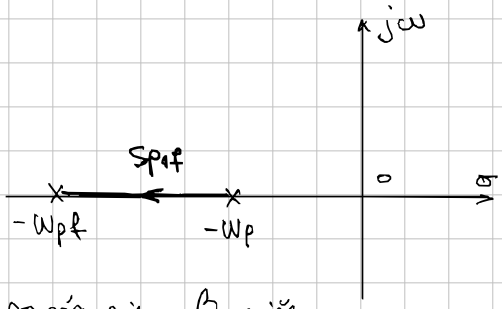
bez P.V.

$$\beta = \beta_0 \neq f(s) \longrightarrow A_f(s) = \frac{A(s)}{1+\beta A(s)} = \frac{A_{0f}}{1+\frac{s}{\omega_{pf}}}$$

sa P.V.

$$\Rightarrow A_{0f} = \frac{A_0}{1+\beta A_0}$$

$$\omega_{pf} = \omega_p(1+\beta A_0)$$



povećavanjem β miče se UJEVO

→ promjenom P.V. pol ostaje u neg. poluravnini ⇒ bezuvjetno stabilno

sustav s 1 polom = nemoguće za napraviti

Pojačalo s dva pola u prijenos. fji

$$A(s) = \frac{A_{0f}}{\left(1+\frac{s}{\omega_{p1}}\right)\left(1+\frac{s}{\omega_{p2}}\right)}, \quad \beta = \beta_0 \neq f(s)$$

POLOVI: $s_{p1} = -\omega_{p1}$, $s_{p2} = -\omega_{p2}$

$$A_f(s) = \frac{A(s)}{1+\beta A(s)} = \frac{A_{0f}}{1+a_1 s + a_2 s^2}$$

$$a_1 = \frac{1}{1+\beta A_0}\left(\frac{1}{\omega_{p1}} + \frac{1}{\omega_{p2}}\right)$$

$$a_2 = \frac{1}{(1+\beta A_0)\omega_{p1}\cdot\omega_{p2}}$$

Drugi oblik:

$$A_{fs} = \frac{A_{0f}}{1+\left(\frac{1}{Q}\right)\left(\frac{s}{\omega_0}\right)+\left(\frac{s}{\omega_0}\right)^2}$$

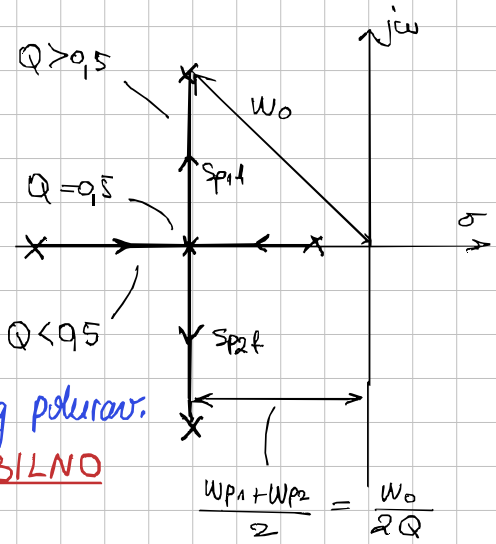
$$\longrightarrow \omega_0 = \sqrt{(1+\beta A_0)\omega_{p1}\cdot\omega_{p2}}$$

$$\longrightarrow Q = \frac{\omega_0}{\omega_{p1}+\omega_{p2}}$$

Polovi: $\left(\frac{s}{\omega_0}\right)^2 + \left(\frac{1}{Q}\right)\left(\frac{s}{\omega_0}\right) + 1 = 0$

$$s_{p1,2f} = \frac{-\omega_0}{2Q} \pm \frac{\omega_0}{2Q}\sqrt{1-4Q^2}$$

$$= -\frac{\omega_{p1}+\omega_{p2}}{2} \pm \frac{\omega_{p1}+\omega_{p2}}{2}\sqrt{1-4Q^2}$$



Primjenom P.V. polovi ostaju u neg polurav.
↳ pojačalo BEZUVJETNO STABILNO

- konjugirano-kompleksni polovi ⇒ uzrokuju prijelazne oscilacije

dalje primjer polova gledaj u ppt

Zadatak 1.) Poj s P.V. $A(s) = \left(\frac{10}{1 + \frac{s}{10^4}} \right)^3$

polovi? dijagram? $\beta \rightarrow$ nestabilno?

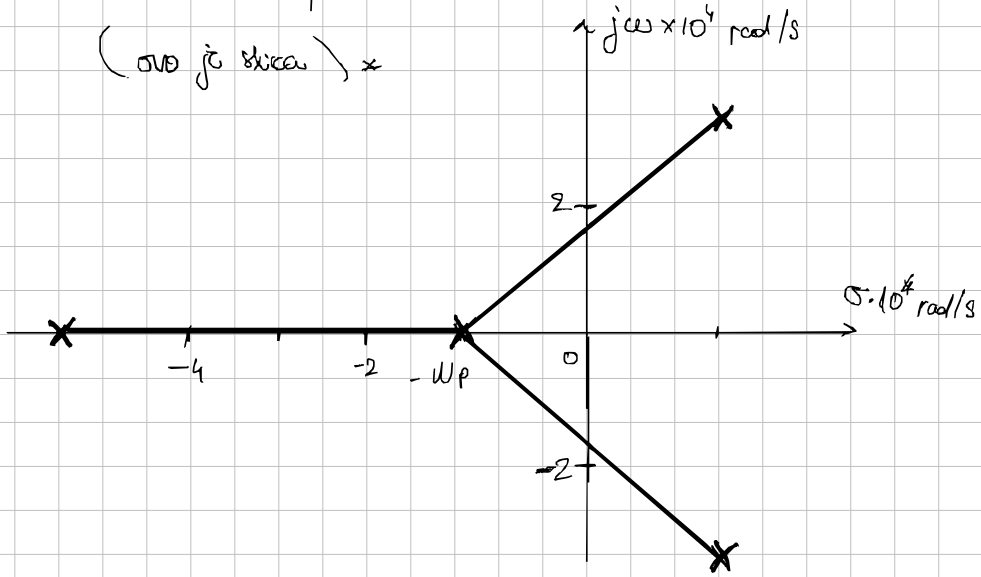
$$A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_p}\right)^3} = \frac{10^3}{\left(1 + \frac{s}{\omega_p}\right)^3} \qquad \omega_p = 10^4$$

$$1 + \beta A(s) = 1 + \frac{10^3 \beta}{\left(1 + \frac{s}{\omega_p}\right)^3} = 0$$

NULE => $\left(1 + \frac{s}{\omega_p}\right)^3 + 10^3 \beta = 0$

$$\left(\frac{s}{\omega_p}\right)^3 + 3\left(\frac{s}{\omega_p}\right)^2 + 3\left(\frac{s}{\omega_p}\right) + 1 + 10^3 \beta = 0$$

$$s_{pe_{1,2}} = -\omega_p \left(1 - 5\beta^{1/3} \mp j5\sqrt{3}\beta^{1/3}\right)$$

$$s_{pf_3} = -\omega_p \left(1 + 10\beta^{1/3}\right)$$

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